

## What's Happening Around The World

1. The United States alone has approved \$52 billion in spending for domestic semiconductor manufacturing and research.
2. China has sanctioned more than 40 billion yuan (\$6.19 bn) in combined funding during the first five months of 2021 to more than a 100 semiconductor companies.
3. South Korea has plans to spend roughly \$450 billion to build the world's biggest chipmaking base over the next decade.
4. Japan is said to be offering a total of several hundred billion yen, equivalent to several billion dollars, to overseas chipmakers ready to build advanced chip manufacturing plants in the country.
5. The government of India is about to launch a new scheme for attracting global semiconductor companies to set shop in the country.

tinues to spread over different parts of the world, electronics manufacturers will want to phase out existing devices based on 4G tech, and the process will require sourcing more and better semiconductors.

A report by Accenture, cementing this notion, has forecast that starting 2023, smartphones with 5G capability will sell more than 4G-compatible smartphones. As against 614 million 4G smartphones forecast to sell in 2023, a total of 995 million 5G handsets have been forecast to sell in that year. These numbers will keep on increasing to 1,143 million 5G handsets in 2024 compared to 470 million 4G handsets same year.

"The 5G has not gone up in volumes in terms of deployment by operators. But 5G demand is going to increase during the next couple of years, and products for it would ramp up in the coming years," says Hemant.

The chip volume and RF frontend modules for 5G and 5G mmWave are expected to double in premium smartphones, and 5G semiconductor revenue—including baseband processors, RF, and power management—will increase from near zero in 2018 to \$31.5 billion in 2023, the vast majority of which will be driven by smartphones.

A report by PMI has forecast that the global 5G semiconductor solutions market would be worth US\$10.8 billion by 2029. It was valued at around US\$1.5 billion in 2020. The growth and development in the ICT sector to improve wireless data networks, communication technologies, and solutions through fifth-generation network usage are expected to fuel the growth of the global 5G semiconductor solutions market.

This simply means that 5G is helping to increase the adoption of the latest tech based semiconductors in the world, and it is a known fact that products based on latest technologies usually cost more than the outdated or soon to be outdated technologies. Replacement of devices compatible with 4G by those compatible with 5G is also leading to an increase in the demand of semiconductors. However, as a lot of fabs are still struggling to update to the latest tech, there continues to be a shortage of the same in the market.

The shares of TSMC, a manufacturing vendor for a lot of fabless semiconductor companies, have delivered 106.87 percent RoI during the last one year. TSMC operates more than one fab based on the latest technologies and it is known as one of the fabs to always be on the top of the technology game. The company is targeting to start volume production using its latest 3nm tech in the second half of 2022.

TSMC had a market cap of \$604.33 billion on 28th June 2021. Its stock prices rose from \$56.33 per share (June 2020) to \$116.53 per share (June 2021).

### Automotive, EVs, and battery industry

A lot of experts and consulting organisations are of the view that the period of 2021 to 2030 will be the decade of electric vehicles (EVs). India and many other countries have already announced aggressive policies to increase the adoption of EVs. The global sales of EVs in 2020, as per a report by Canalys, increased by 39 percent in a year to 3.1 million units. This is when the total passenger car market declined 14 percent.

Canalys has also forecast that these figures will climb to 30 million EVs sold in 2028. EVs are expected to represent nearly half of all passenger cars sold globally by 2030. Now there is no rocket science in knowing that the more sophisticated an electric car model is, the more semiconductors are required to power it.

"EVs being very hot right now, there is already very high demand from the sector. Automotive uses a large number of sensors and electronics now. I think the sector has also contributed to the increased demand of semiconductors," explains Hemant.

Global demand for automotive semiconductor market, in terms of revenue, was worth \$45.98 billion in 2020 and is expected to reach \$81.40 billion in 2027 (Brandessence Market Research). As a matter of fact, even ICE powered vehicles enabled with security features like camera and connectivity require a good number of semiconductors. Moreover, as the autonomous driving feature starts becoming more common, the quantity of semiconductors demanded by automotive OEMs will see an even sharper incline.

Since 2020, automotive OEMs have been ramping up the production of EVs and hence demanding more and more semiconductors from companies. The likes of Audi have already announced to be EV-only brand by 2026.

Batteries and charging points are two other industries that are directly related to the EV ecosystem, and both of these require semiconductors in good numbers. Batteries require semiconductors because developments in the semiconductor arena can help provide longer battery life and range, and charging points because they are all electric.

"With congestion in the public transport sector, a lot of people are investing in buying private vehicles. Remember the amount of electronics in these vehicles. Forget about high-end cars and talk to Maruti and you will be surprised to know the number of electronics items that go into an entry segment car like Alto 800," says Menon.

"Then look at two-wheelers and you will find semiconductors worth at least \$7-8 going into an entry level two-wheeler. The mandating of Bharat Stage-6 emission standard for all vehi-